Remote Sensing of Environment 308 (2024) 114206

Contents lists available at ScienceDirect

Remote Sensing of Environment

journal homepage: www.elsevier.com/locate/rse

Learning the variations in annual spectral-temporal metrics to enhance the transferability of regression models for land cover fraction monitoring

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ARTICLE INFO

Editor: Marie Weiss

Keywords:
Land cover fractions
Regression-based unmixing
Sentinel-2
Spectral-temporal metrics
Data density

ABSTRACT

Monitoring the Earth by annually mapping land cover (LC) fractions helps to better understand the ongoing processes and changes of land use and land management. At 10 to 30 m spatial resolution, the combination of time-series data aggregation, specifically spectral-temporal metrics (STM), and regression-based unmixing models has been shown to be highly effective in quantifying LC fractions over large areas. However, STM are subject to variations in data densities within and between years, which may lead to variations in prediction accuracies and limit the transferability of models through time. To better understand the influence of annual data density on multi-year monitoring, we systematically tested the accuracy and spatial-temporal transferability of regression models for LC fraction mapping. Additionally, we introduced a novel strategy, Random Observation Selection, in the STM generation to enhance model transferability and compared its results to those obtained from regular STM. We used STM from multi-spectral Sentinel-2 satellite data to estimate Impervious surface, Woody and Low vegetation at a regional scale in northern Germany and Poland for the years 2017 to 2022. The study period is characterized by substantial inter-annual variation in data density and well suited to test temporal transfer: in the extremely hot and dry years of 2018 and 2019 data density is above average and considered relatively high, while in the wetter years of 2017 and 2021 it is low. The mapping quality of models trained in single years varied considerably between years with high and low data density. Specifically, models trained in years with high data density performed well in that same year but poorly when transferred to years with lower density. Conversely, models trained in years with low data density demonstrated less of a decrease in accuracy when transferred. A multi-year model trained in all six years performs the best for each individual year. The use of Random Observation Selection improved the transferability of all models, particularly for those trained with data from a high data density year when transferred to years with lower data density. Here, the mean absolute error showed the highest relative improvement of 16%. In conclusion, STM proved useful for multi-year regression-based monitoring, but care must be taken when annual data density varies. Incorporating Random Observation Selection reduced this influence and improved the spatial-temporal transferability of quantitative LC fraction monitoring.

1. Introduction

Earth observation (EO) offers unique opportunities to explore the dynamics of land cover (LC) and, this way, delivers insights into land use and land management (Pongratz et al., 2018; Woodcock et al., 2020). A better understanding of the underlying processes of land use and management requires time-series of quantitative information such as maps of sub-pixel LC fractions that represent both abrupt changes (Lunetta et al., 2006; Frazier and Wang, 2011) and gradual dynamics (Vogelmann et al., 2016; Potapov et al., 2019; Chen et al., 2021; Stanimirova et al., 2022). Over the past ten years, LC fraction maps from finer resolution data (10–30 m) have increasingly been used for

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https://doi.org/10.1016/j.rse.2024.114206
Received 18 September 2023; Received in revised form 13 May 2024; Accepted 13 May 2024
Available online 16 May 2024
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monitoring gradual dynamics and disturbances through space and time for various ecosystems, e.g., grasslands (Lewińska et al., 2020), shrublands (Suess et al., 2018), forests (Hansen et al., 2013; Buchhorn et al., 2020) or urban areas (Lefebvre et al., 2016; Wellmann et al., 2020; Farhadi et al., 2023). In this context, machine learning based regression is increasingly used for directly estimating LC factions from the full spectral information of EO data (Deng et al., 2017; Baumann et al., 2018; Nill et al., 2022; Viana-Soto et al., 2022).

For the future, a more operational usage of time series and applications over larger areas can be expected (Woodcock et al., 2020). Large area monitoring is ideally conducted in annual or even intra-annual intervals to capture all related processes (Hansen and Loveland, 2012). When mapping areas larger than single orbit frames at 10–30 m resolution, i.e., with Sentinel-2 or Landsat sensors, image acquisition dates in pixel-based composites will always vary depending on the methodology and can lead to unnatural offsets in LC fraction predictions during mapping (Xian et al., 2015). Orbit constellations and gaps caused by cloud coverage lead to differences in data availability for Landsat or Sentinel-2 sensors throughout space and time, especially in mid-latitude, temperate regions (Griffiths et al., 2013). For example, such variations in clear sky observations (CSO) have been shown to have great impact on the reconstruction of Landsat time series information (Pouliot and Latifovic, 2018; Zhang et al., 2021). Annual metrics of spectral variations, also termed spectral-temporal metrics (STM, Frantz, 2019), are pixel-wise statistics (e.g., the annual minimum, maximum, median, or quartiles) calculated per band (or per spectral index) from all available observations in a pre-defined time period which are often continuously available across space. While STM are still sensitive to variations from acquisition differences or phenology (Frantz et al., 2023), they are considered to be less prone to such effects than compositing approaches and they are expected to generate more homogeneous inputs for analyses and still represent intra-annual signal variations (Hansen et al., 2013; Baumann et al., 2018). In recent years, STM have been commonly used in combination with machine learning models for mapping discrete LC classes (Pasquarella et al., 2018; Pflugmacher et al., 2019; Zhang et al., 2023). Applications of STM for quantitative mapping have also been adopted widely, for example, Hansen et al. (2013) used annual Landsat STM of four fraction intervals in their seminal work on global forest cover between 2000 and 2012. Jones et al. (2018) mapped annual LC fractions (forbs, grasses, shrubs) for the western US from 1984 to 2017 using annual STM from Landsat data. It should be noted that given the sparsity of temporal data, it is often necessary to aggregate a longer time-series of data to ensure mapping consistency. For example, Nill et al. (2022) aggregated observations for 6-year intervals before calculating STM to handle sparse data, when mapping arctic shrub expansion from Landsat STM between 1984 and 2020. In another study, Schug et al. (2020) aggregated 2-year (2017–2020) Sentinel-2 A/B data to map urban-rural LC fraction gradients with 10 m resolution for Germany and Austria. While Sentinel-2 data has a higher revisit frequency compared to Landsat data, the inter-annual variations in data density and distribution still affect STM values in multi-year time series substantially (Fig. 1).

The impact of these STM variations on model performances has yet to be assessed for annual monitoring of LC fractions. However, it would be beneficial for supervised methods, e.g., the machine learning regressions, to consider these variations when monitoring LC fractions over multiple years from annual STM. Employing a new model every year to accommodate the data would be highly duplicative and computationally costly, in comparison to using a single generalized model (Woodcock et al., 2001). Various studies have shown that a single machine learning regression model is capable of learning information from multiple years, i.e., leading to a single generalized multi-year model (Jones et al., 2018; Suess et al., 2018; Stanimirova et al., 2022). However, these studies used different strategies for multi-year model training, which heavily depended on the availability of reference information. For example, Jones et al. (2018) used quantitative information from >27,000 plots from field surveys between 2006 and 2016 to train and validate their regression-based results for the western US. For many regions in the world, however, such detailed information will not be available for all years within a monitoring period. Alternatively, co-registered very high resolution (VHR) imagery at improved spatial resolution may be digitized or classified to create reference information for regression-based LC fraction mapping (Xian et al., 2015). Still, the creation of annual reference information can be laborious, it is prone to geometrical uncertainty and relies on multitemporal VHR data.

To make the creation of regression training data less dependent on reference information and easier to gather, Okujeni et al. (2013) suggested the use of spectral libraries comprising spectral signatures for singular LC types, e.g. labelled and extracted from “pure image pixels,” similar to the generation of training information for supervised classification. Synthetic spectra are created by mixing two or more LC of the same or different classes. The ratios of the mixture serve as target variables in regression training, with the mixed spectra used as input variables (Okujeni et al., 2013; Okujeni et al., 2017). This regression-based unmixing with synthetic training data has been shown to be less tedious in reference data generation, yet similar in terms of accuracy compared to training based on human-annotated sub-pixel fractions (Priem et al., 2019). The regression-based unmixing approach was used by Suess et al. (2018) to map 32 years of shrub cover with Landsat composites data. A similar approach was then adopted from Senf et al. (2020) to map forest cover fraction over three decades. In recent studies, Viana-Soto et al. (2022) mapped 30 years of post-fire vegetation dynamics for two study sites in Spain based on a time-series of Landsat STM and a single multi-year regression model. All these studies shared the concept of creating a regression model that generalizes over extended time periods by expanding reference information from a single year over time. However, this expansion depends on the possibility to verify the temporal invariance of training pixels, and neither examined the impact of annual data availability on results from regression modelling, particularly when using annual STM as input feature. It is important to understand this

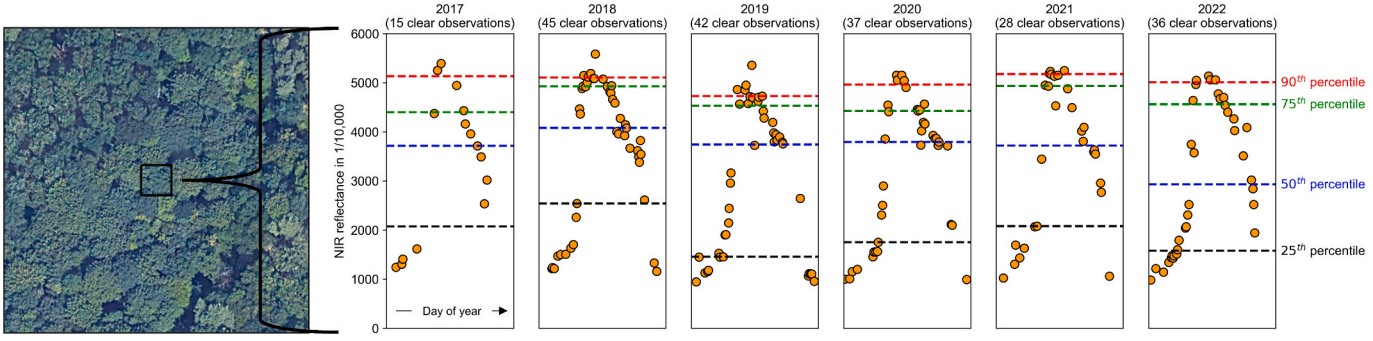

Fig. 1. 6-year time series (2017–2022) of Sentinel-2 A/B near-infrared values (Band 8) from a deciduous forest pixel (10 m resolution) in northern Germany together with annual STM. The differences in annual STM highlight the influence of data density and distribution.

influence when mapping consistency is needed to derive gradual trends reliably. Ideally, model training should then be exposed to the spectral variations regarding the annual differences in observations. Thus, training with single-year reference information regarding possible annual variations over multiple years may make such consistency checks obsolete, which is desirable. Augmentation techniques have proven successful in introducing variations during model training, such as for time series classification. For example, Graesser et al. (2022) created pseudo temporal sequences by sampling LC change within a single time step. In another study, Geiß et al. (2019) introduced artificially transformed training samples to the classification model to improve LC mapping accuracy. For the work with STM time series, the augmentation of STM expressions that represent temporal sub-samples of actual observation sequences appears feasible, because no new information must be searched or created. In this case, the augmentation would incorporate information representing an artificially decreased data density.

In this study, we examined the influence of temporal data density on the estimation of LC fractions using regression-based unmixing and annual Sentinel-2 STM data. We conducted experiments over an extensive area covering the southern Baltic Sea region for six consecutive years (2017–2022) at 10 m resolution. Our study focused on three LC classes: *Impervious surface* (comprising built-up and artificial infrastructures), *Woody* (encompassing trees), and *Low Vegetation* (non-woody vegetation including crops, grassland, herbaceous and wetland vegetation). We focused on the following research questions for this setup:

1. To what extent does the temporal data density influence the accuracy of regression models during temporal and spatial transfer?
2. Can model transfer be improved by artificially incorporating the variation of temporal data density during model training?
3. Does annual LC fraction mapping allow for accurate mapping over multiple years?

To answer these questions, we first trained single-year and multi-year regression models using a synthetic unmixing approach and assessed the performance of each model when applied to its respective year. Subsequently, we evaluated the transferability of the model when applied to other years, with additional spatial transferability being investigated. Based on these insights, we developed and tested a new approach for enhancing the temporal transfer of the regression-based unmixing models, denoted as Random Observation Selection (ROS). We compared results from strategies with and without ROS. To evaluate the approach for multi-year monitoring, we selected the best generalizing model (i.e., model performed the best in 6-year mapping) to generate LC fraction maps for the entire area. Results were visually and quantitatively assessed regarding the consistency of invariant LC, as well as gradual and abrupt LC fluctuations throughout the research period on areas undergoing changes.

To organize our study, we have described our methods and workflows in Section 2. This includes the description of the study area and image data, as well as an explanation of our experiments regarding regression-based unmixing approaches and the proposed ROS method. Section 3 provides detailed results of single-year and multi-year trained models, with and without ROS. We also show the visual assessment of annual LC fraction maps from the overall best model. Section 4 discusses the related results and recommendations, followed by the conclusion in Section 5.

2. Materials and method

2.1. Study area and Sentinel-2 data

The study area is located in the southern Baltic Sea region including northern Germany and Poland, i.e., 405,000 km^2 covered by seven Sentinel-2 orbits. The area comprises a mix of various LC classes that occur in regionally different spatial patterns, e.g., larger cropland parcels in former eastern Germany or western Poland and smaller parcels in the former western Germany or eastern Poland, or urban agglomerations of different density. Sentinel-2 A/B data availability in terms of annual CSO from 2017 to 2022 varied substantially (Fig. 2). In 2017 and 2021, extended rainy and cloudy periods contributed to a decrease in CSO. Additionally, the low data density in 2017 (average per-pixel CSO: 13 ± 4) was due to the late onset of Sentinel-2B's operational phase that year. Conversely, high data density was found in 2019, 2020, 2022 (average per-pixel CSO: 30 ± 8), and particularly in 2018 with an extremely dry and hot summer (average per-pixel CSO: 35 ± 9). In addition to year-to-year variation, the overlap of orbits generated a spatial arrangement that resulted in varying data density across space. In summary, the study area and observation period represent a challenging case for investigating LC fraction monitoring of a temperate region.

We acquired all available Sentinel-2 A/B Level-1C images with cloud cover below 75% for 2017 to 2022 provided by the European Space Agency (ESA). We used this threshold because the share of unusable data is higher than indicated by the cloud coverage provided in the Sentinel-2 metadata. Metadata cloud cover estimates do not include cloud shadows, cirrus clouds and cloud buffer areas, leading to many unusable pixels and little discarded usable data if images with a metadata cloud cover >75% are excluded. Data pre-processing was performed in the open-source *Framework for Operational Radiometric Correction for Environmental monitoring* – FORCE (Frantz, 2019). FORCE generates a tile-based structure for remote sensing data (in our study area, each tile covers 30 × 30 km^2) that allows a parallel workflow for mapping large areas. In our study area, all Sentinel-2 data were processed in a total of 450 tiles (Fig. 2). FORCE produces Sentinel-2 Level 2 Analysis Ready Data (ARD) by performing radiometric correction following (Frantz et al., 2016a) and cloud masking according Frantz et al. (2018) based on the Fmask algorithm (Zhu and Woodcock, 2012; Zhu et al., 2015) In this study, we used 10 spectral bands of Sentinel-2, i.e., the four 10 m bands (2, 3, 4, 8) and six 20 m bands (5, 6, 7, 8 A, 10, 11), which were resampled to 10 m resolution using the *Improving the spatial resolution of land surface Phenology algorithm* – ImproPhe (Frantz et al., 2016b). For the later mapping with the different regression models, the following annual STM were calculated within FORCE for all six years of the study period and all spectral bands: 25th, 50th, 75th, 90th, percentiles. The combination of these metrics was found to achieve high accuracies in regression-based unmixing of a similar area for the years 2017/2018 (Schug et al., 2020).

2.2. Creation of spectral libraries

Regression-based unmixing using synthetic training data requires spectral libraries derived from pure LC surfaces (Okujeni et al., 2013). For spectral library creation, we selected five tiles located within Germany (Fig. 2), which include dense urban areas, as well as agricultural and forest areas. Within these regions, we identified pixels solely made up of LC from one of the target classes of *Impervious surface*, *Woody* or *Low vegetation.* We deliberately gathered training data exclusively in Germany, omitting Poland, to assess the spatial transferability of models among considerably disjointed areas within our study region. For each LC type we employed different strategies to collect the pure spectra for synthetic mixing of training data:

- *Impervious surface:* Pure pixels of *Impervious surfaces* were initially selected by interpreting high-resolution imagery from Google Earth combined with the Imperviousness raster dataset (2018, 10 m resolution) from The Copernicus Land Monitoring Service (CLMS, https://land.copernicus.eu/en/products/high-resolution-layer-imperviousness). From Google Earth imagery, we manually selected pixels that have LC homogeneity (at least 30 × 30 m^2 homogeneous) and consistency (no change during 2017–2022). From the initial selection, only pixels with 100% imperviousness from CLMS dataset were used further. Next, from

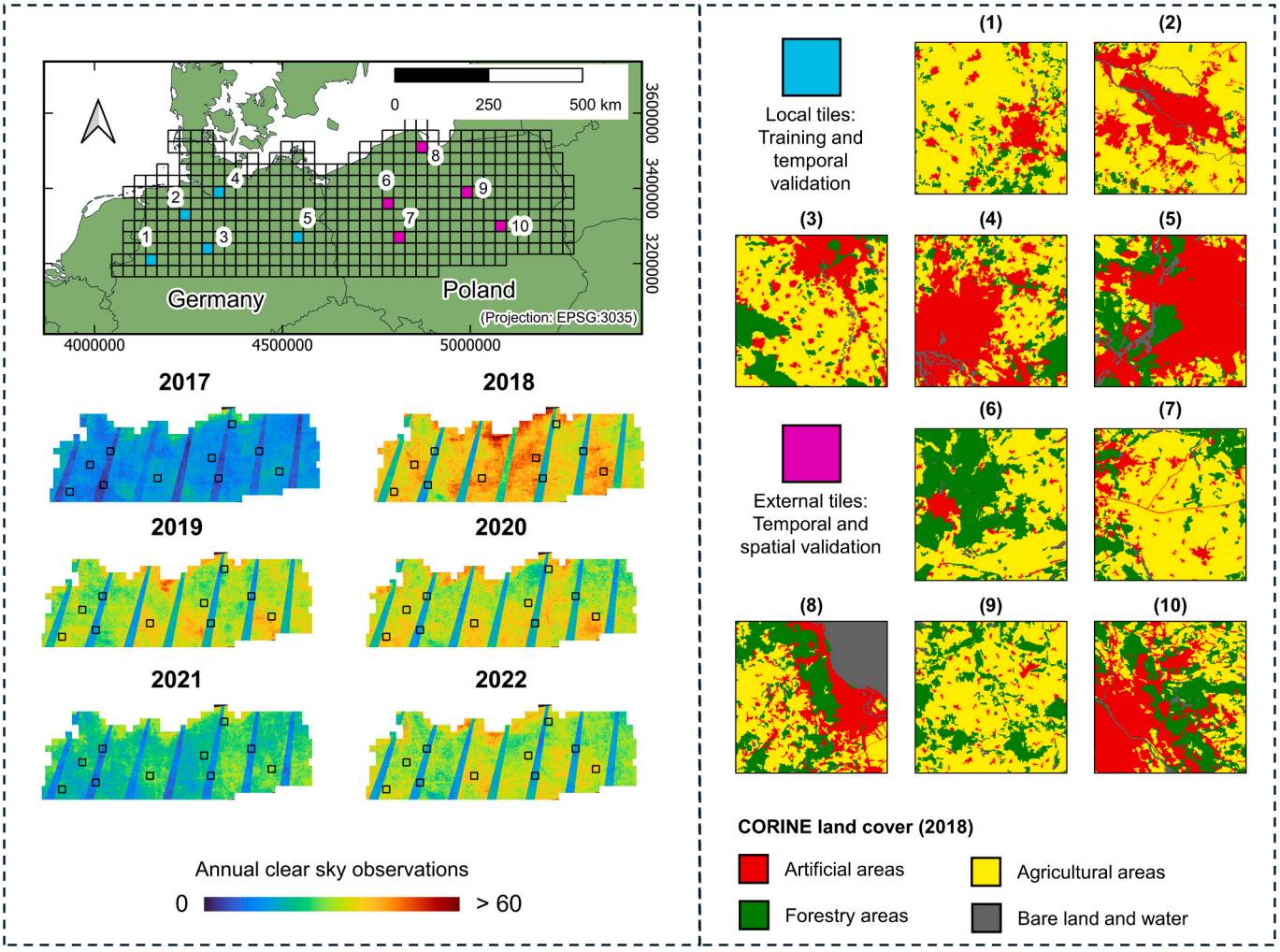

Fig. 2. Study area extent in northern Germany and Poland and annual temporal density of clear sky observations of Sentinel-2 A/B from 2017 to 2022 (left). Locations of local tiles (1–5, for training and temporal validation) and external tiles (6–10, for temporal and spatial validation), with corresponding LC information according to CORINE Land Cover (2018, https://land.copernicus.eu/en/products/corine-land-cover; right).

the selected pixels, we calculated the 90th percentile *Normalized Difference Vegetation Index* – NDVI (Tucker, 1979) within a 6-year period (2017–2022). We then removed pixels that have 90th NDVI values higher than 0.1. This practice is inspired from previous studies (using maximum NDVI >0.3 threshold) to eliminate pixels that potentially contain small vegetation objects (Liu and Juárez, 2001; Esau et al., 2016; Schug et al., 2020).

- *Woody*: We first used the Tree Cover Density raster dataset (2018, 10 m resolution) from CLMS (https://land.copernicus.eu/en/products/high-resolution-layer-tree-cover-density) to identify potential pure woody pixels (tree density values = 100%), most of which are located in the middle of forest areas. Subsequently, we used the 90th percentile NDVI value within a 6–year period to select reference pixels with values larger than 0.8, as this measurement was used to identify pure dense forest (Asrar et al., 1984). Finally, for each chosen pixel, we manually inspected the consistency with historical VHR from Google Earth Pro in order to avoid disturbances (e.g., deforestation, forest fire, beetle infections) during 2017–2022.

- *Low-vegetation* refers to non-woody vegetation cover including grasslands, herbaceous, wetland vegetation and crops, the latter potentially including open soil for parts of the year. As a start, we used the CORINE Land Cover raster layer (2018, 10 m resolution, https://land.copernicus.eu/en/products/corine-land-cover/clc2018) to identify the potential locations of these cover types. Then, pixels located in the center of homogeneous surfaces (e.g., in the middle of crop parcels) were selected. A final manual check on LC homogeneity and consistency was performed using time-series VHR from Google Earth Pro.

- Background class: the three target classes may be mixed with additional LC types, e.g., permanent soil, bare rock, permanent water (all are aggregated in one class), referred to as background class in the approach of Okujeni et al. (2017). Pure pixels for this class were initially selected using VHR imagery. Similar to the method selecting pure *Impervious surface*, to avoid pixels that might contain small vegetation objects, we also applied a threshold using 90th percentile NDVI calculated from 2017 to 2022 and removed pixels exceeding the threshold of 0.1.

In total we sampled 966 reference pixels for the four classes (Table 1). Subsequently, the full Sentinel-2 A/B time series of clear observations was extracted for all sampled pixels, and we calculated annual STM for the six investigated years and stored them in separate annual libraries.

When the STM from the four chosen aggregation functions (25th, 50th, 75th, 90th percentiles) are represented as spectral profiles, the temporal variations within each plot become obvious (Fig. 3). Profiles generated on *Impervious surfaces* are invariant within and between years, because such surfaces have no plant phenology and do not vary during drier or wetter years. Annual STM profiles represent typical inter- and intra-annual variations of deciduous woody vegetation very well. The

Table 1
Number of pure pixels sampled per target-class and sub-classes.

Target class	Sub-classes	Number of pure pixels
Impervious surface	Dark roof building	65
(Total = 298)	Bright roof building	75
	Railways	77
	Roads	72
Woody	Coniferous forest	78
(Total = 208)	Broadleaved forest	130
Low vegetation	Cropland	150
(Total = 362)	Grassland	147
	Wetland vegetation	65
Background	Water surface	64
(Total = 98)	Permanent soil	34

relatively long green period (wherein observations are also most likely to occur) leads to little difference between the 90th and 75th percentile (NIR band - wavelength 842 nm), while the 25th percentile deviates more, because it is more influenced by irregular and sparse acquisitions from late fall to early spring. STM profiles from cropland, as an example of low vegetation, are influenced by both vegetation and soil cover. Thus, the 90th percentile describes a high NIR reflectance from periods with high vegetation cover, as well as red reflectance influenced by periods of soil cover and, hence, higher than for dense vegetation. The more pronounced phenology on these mostly managed sites differs more clearly between years regarding the absolute values of STM curves and the differences between singular metrics within years.

2.3. Regression-based unmixing using synthetic training data

Regression-based unmixing approaches have so far used a variety of machine learning algorithms, e.g., Kernel Ridge Regression, Random Forest or Support Vector Regression (Campos-Taberner et al., 2016; Lopatin et al., 2016; Senf et al., 2020). In this study, we developed a workflow based on multi-class Neural Network Regression (NNR). This is because NNR allows multi-class regression in one single prediction (Specht, 1991) and hence requires fewer parameters compared to multiple single-output regressors (Garioud et al., 2021). Furthermore, during training, the NNR model is optimized by a multi-target loss function, enabling the model to learn the dependencies between all target classes (Heiat, 2002). This potentially facilitates multiple fractions of single prediction to sum up to unity (i.e., 100% fraction cover) and, this way, physically more meaningful predictions. We set up the NNR architecture consisting of two hidden layers (512 nodes each) that take an input of 40 parameters (10 bands × 4 metrics) and output four probabilities, which are corresponding to four LC fraction predictions (*Impervious surface*, *Woody, Low vegetation*, *Others*). We used Mean Absolute Error as loss function to train the model, a learning rate of 0.001 and the Adam optimizer (Kingma and Ba, 2014) for optimizing the weights of the NNR. The process was implemented using the Tensorflow Python framework version 2.10.0 (Martín et al., 2015).

We incorporated the concept of synthetic mixing, as inspired by recent studies (Priem et al., 2019; Cooper et al., 2020; Okujeni et al., 2021), which initially focused on single-output regression. In this study, we extended this approach to multi-output regression, as illustrated in Fig. 4. First, we derived STM from pixels of pure LC without change from 2017 to 2022 (Fig. 4a). Synthetic training data were generated by randomly mixing pairs and triplets of pure STM from the same year in varying ratios (Fig. 4b). This process involves the following steps: First, we randomly selected a certain number (ranging from 1 to 4) from a pool of pure STM for a specific year. Next, based on the selected number of pure STM and their corresponding LC types (the same LC can be drawn in one selection), we created a set of four arbitrary fractions representing *Impervious surface, Woody, Low Vegetation,* and *Others*. If a particular LC type is not included in the selection list, a corresponding fraction of 0 will be assigned to it. Random fractions between 0 and 1 are generated for the chosen LC, and the sums of all random fractions in one training sample equal 1. Finally, synthetically mixed STM data were generated by summing pure STM weighted by their fractions. The synthetically mixed STM data are used as input and the mixing fractions as target variables to train the Neural Network regression models (Fig. 4c). For each year of the experiment, we created a total of 128,000 pairs of mixed spectra and labels to train the model.

For each year from 2017 to 2022, we trained individual regression models, i.e., *M-17, M-18, M-19, M-20, M-21, M-22*. Additionally, we trained one multi-year model combining the training data from all years (*M-All*). In total, seven regression models were created from the STM

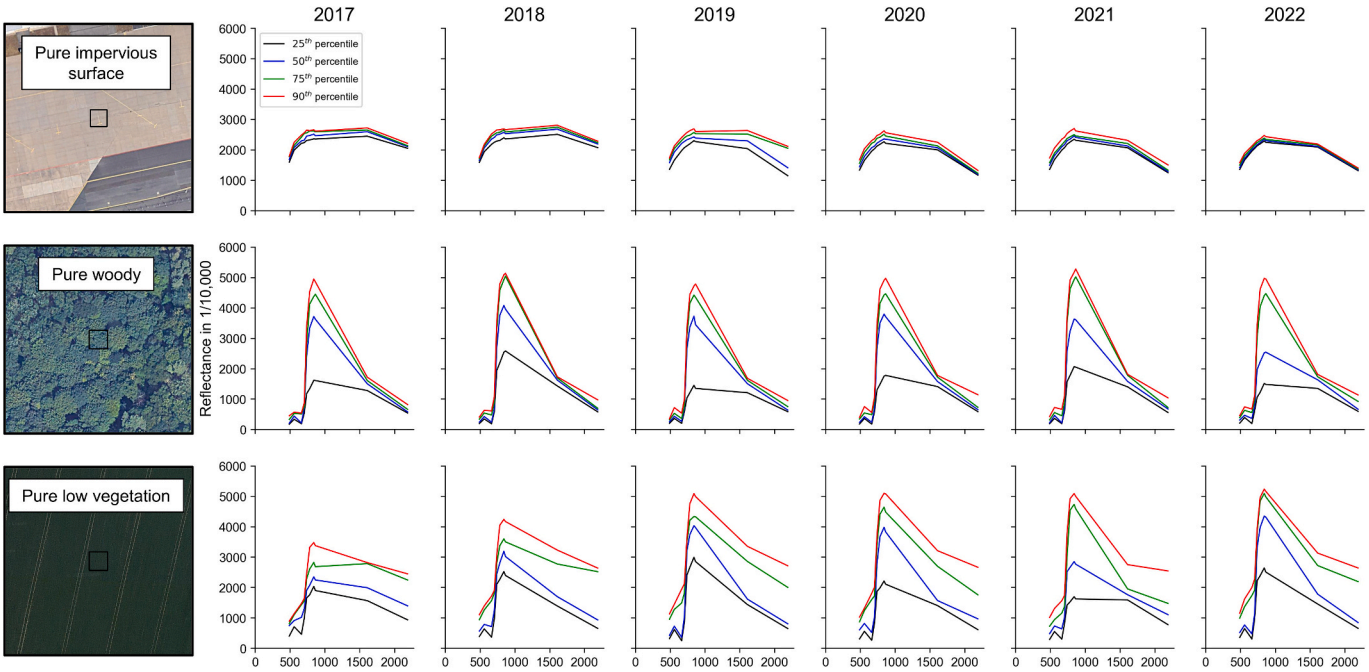

Fig. 3. HR images of pure pixels in three main LC classes: *Impervious surface* (airport runway), *Woody* (broadleaved forest), *Low vegetation* (cropland) together with the respective annual STM profiles (25th, 50th, 75th, 90th percentiles).

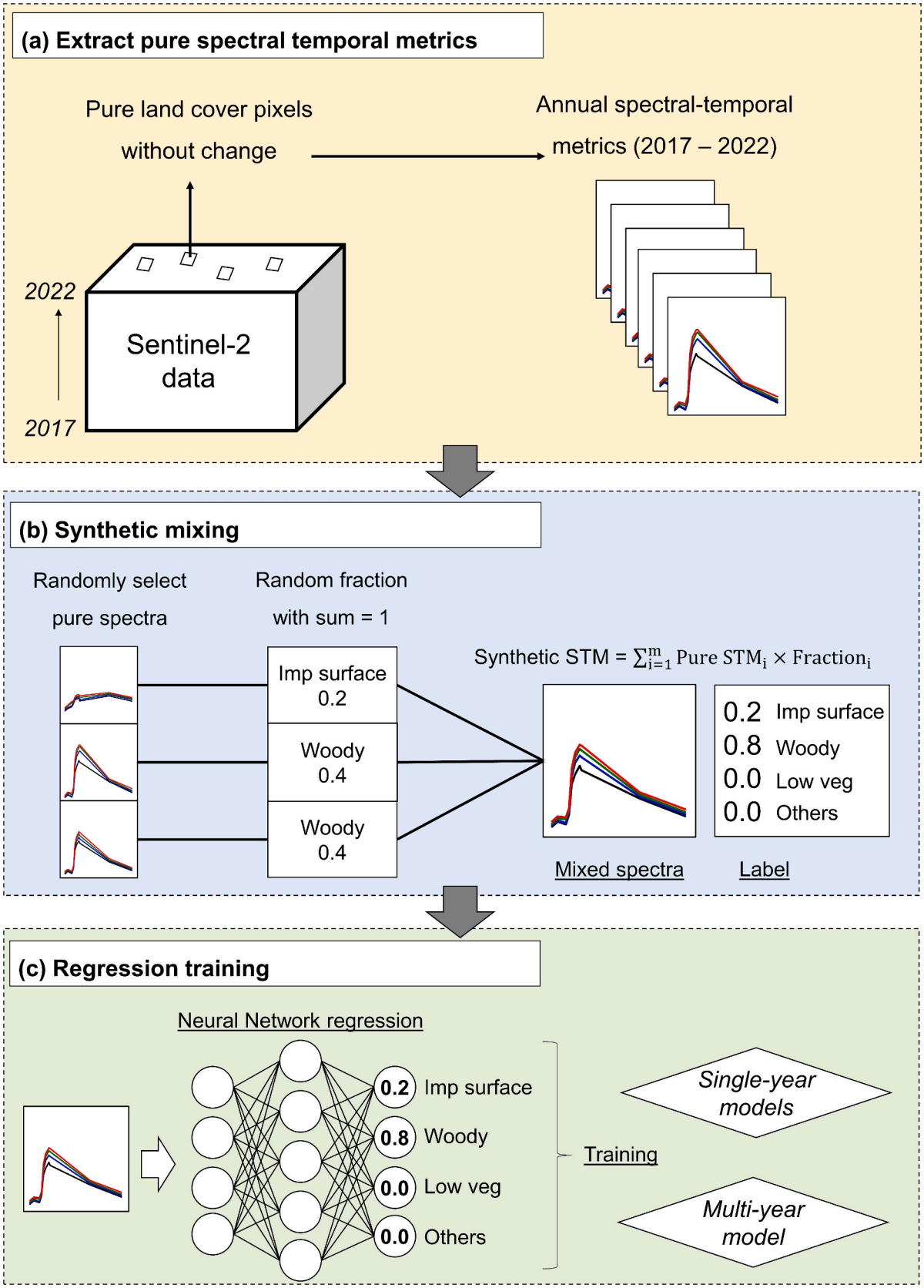

Fig. 4. Workflow of model training with synthetic training data: (a) Pixels from temporally invariant pure LC were sampled. For each sampled pixel, we calculated annual STM based on all clear Sentinel-2 A/B observations; (b) Synthetically mixed STM were generated by mixing multiple pure STM with random fractions; (c) The synthetic STM and fractions are then used as input and label values to train the Neural Network regressions. Single-year models were trained with the subset of synthetic data from the respective year and the multi-year model was trained with all subsets of synthetic data from all years.

libraries of the six years.

2.4. Creating augmented STM using random observation selection

In the standard approach, a single reference pixel of each year yields an annual STM that is utilized for mixing. To enhance the spectral library with STM reflecting various data density and distributions potentially found in regions or years not covered by a specific model, and thus improve the model's transferability, we generated augmented STM for all years (in conjunction with Fig. 4a). From the original annual STM, we derived several instances of STM using reduced portions of all available CSO for that year. We will refer to the resulting set of STM as the augmented STM from now on.

The workflow of creating augmented STM was as follows (see Fig. 5):

- For each year separately, we created random selections from the available CSO of the reference pixels including 80%, 60%, 40%, and 20% of the original count. Therefore, we refer to this procedure as ROS. We kept the minimum CSO of four observations after applying ROS for the training sample. This is suggested by previous studies that the number of CSO should be greater than the number of STM (Potapov et al., 2012; Zhang et al., 2023).

- Next, we calculated the STM (25th, 50th, 75th, 90th percentiles) for each subset of CSO (80%, 60%, 40%, 20%) independently.

As a result, five different sets of STM (including the original and four additional STM derived from ROS) existed for each reference pixel per year. For ROS-enhanced models (i.e., *M-17-R, M-18-R, …, M-All-R*), we combined all sets of original STM and augmented STM to the spectral library for the single-year models and the multi-year model. We used ROS for the training process only. For inference, validation, and mapping, only the original STM were used as input data.

2.5. Validation

In this study, ten tiles were chosen from the entire study area for validation data collection (Fig. 2). The selected validation tiles are situated in the diverse landscapes, i.e., densely populated urban areas, such as major cities like Berlin, Germany and Warsaw, Poland, with a significant proportion of mixed LC pixels. They also encompass more uniform surfaces, such as dense forests and extensive agricultural regions. In addition, we intentionally selected half of the tiles that were situated mostly under single-orbit coverage of Sentinel-2, which often have less CSO compared to other areas in the same year. This practice is to ensure equal validation samples from areas covered by both single and two orbits. Among the selected tiles, five tiles located in Germany (1–5) corresponded to the ones utilized for training data creation ("local tiles"), and the remaining five were independent from the training areas and located in the eastern part ("external tiles"). This approach ensures that validation results from local tiles are not influenced by regional

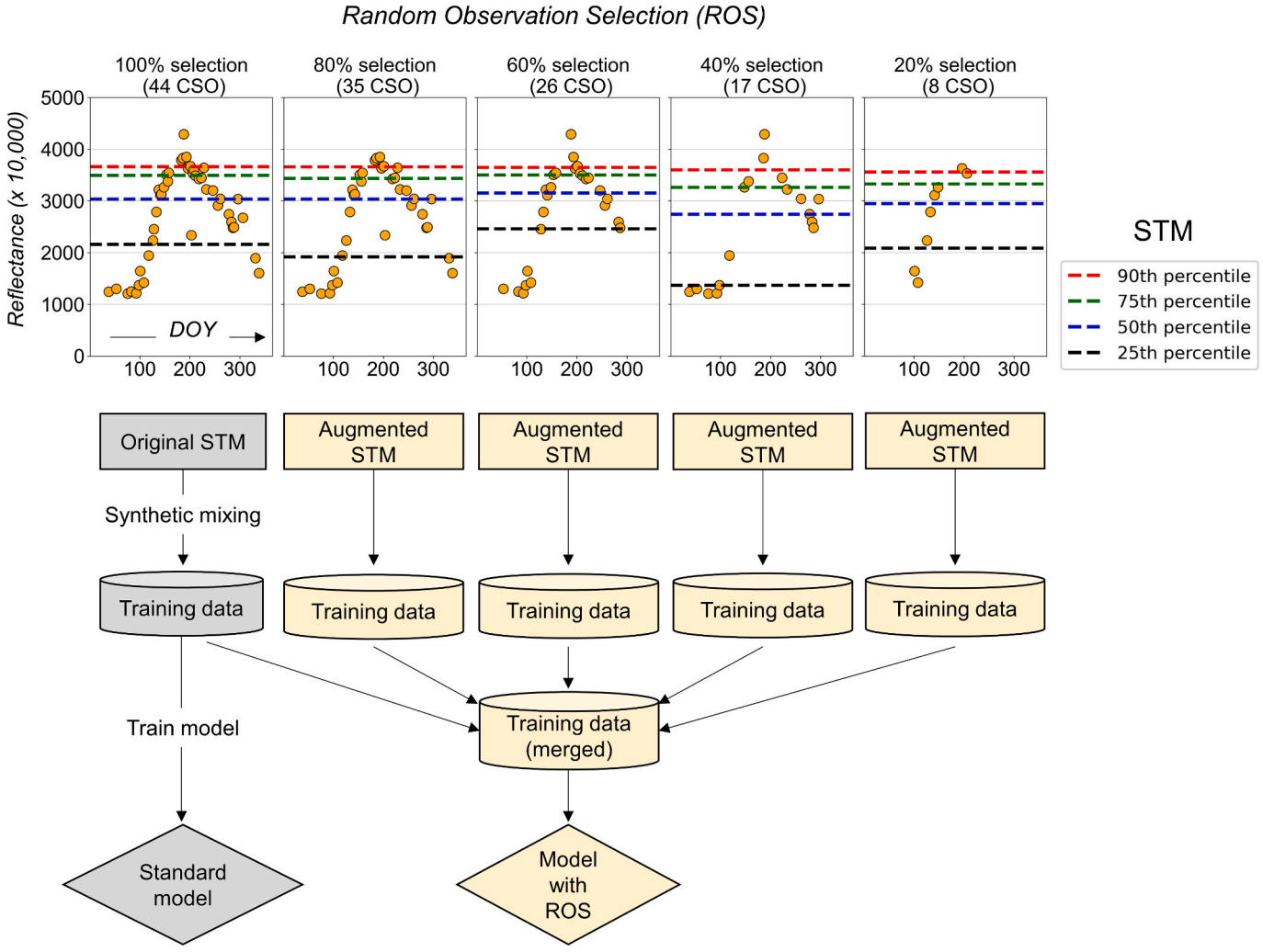

Fig. 5. Illustration of the Random Observation Selection (ROS) and subsequent library generation for model training. The example above shows a time-series of near Infrared reflectance of a pure woody pixel in 2018. The original number is 44 CSO. Original STM values are derived from all available observations. If only a portion of the observations is randomly selected (i.e., 80%, 60%, 40%, 20%), this results in new STM values ("augmented STM").

variation of LC. For external tiles, we intentionally selected more challenging tiles for validations. For example, mixed pixels are often high along urban-rural gradients, hence we included two major cities (Warsaw, Gdansk) plus at least one medium to minor size city in the other three tiles. Three out of five external tiles were situated under a single orbit. Consequently, it emphasizes the effects of temporal transferability (mapping in the same area but in a different year). The results from external tiles, on the other hand, emphasize both temporal and spatial transferability (mapping in a different area and year).

For validation data generation, we followed the established strategy for fraction mapping from previous studies (Schug et al., 2020; Okujeni et al., 2021). First, for each validation sample, LC reference fractions were estimated based on an area of 3 × 3 pixels (30 × 30m). This approach was used instead of validating single pixels due to the geolocation error of Sentinel-2 data, which ranges from 10 to 12 m (Gascon et al., 2017; Rufin et al., 2021) and to decrease the discrepancy between the mapped results and VHR reference data when labelling. Next, we followed a random sampling approach (Olofsson et al., 2014) to stratify validation points from the LC fraction mapping using the intervals 0–20%, 20–40%, 40–60%, 60–80%, and 80–100% (based on the latest LC fraction map - 2022). At each validation point, we overlaid a 10 × 10 cell grid with a resolution of 3 m (Fig. 6a) and manually annotated LC classes using VHR images. Similar to the training data setup, validation sites had to remain invariant in LC fraction over the six-year period to provide a consistent validation framework. This was accomplished by manually examining the pixels for LC change between 2017 and 2022 using historical VHR imagery from Google Earth Pro. Lastly, the reference LC fraction in each class is calculated based on the ratio of their corresponding labelled cells. Due to the inconsistencies in VHR availability as well as geometric miss-alignment, sampling validation points for change areas is hardly achievable in our study. Thus, for change areas, we performed quantitative and visual inspection (see Section 3.5) to evaluate the model's performance.

As a result, we annotated consistent reference fractions (during 2017–2022) for 527 sites within local tiles and 473 sites within external tiles (1000 validation sites in total). It is noted that each validation sample contains reference fractions of all LC classes (sum fraction to 100). The validation data had high shares of mixed fractions (Fig. 6b) in both local and external tiles, hence they are suitable for assessing the sub-pixel mapping results. We then evaluated fraction mapping results for each LC class (*Impervious surface, Woody, Low vegetation*) for the commonly used metrics: Mean Absolute Error (MAE), Root Mean Square Error (RMSE), Slope, Intercept, R^2, as well as visually with scatter plots following the recommendation from Piñeiro et al. (2008).

3. Results

3.1. Performances of annual standard models to the same year

We assessed the performance of all single-year models (standard models) when applied to the same year. We aggregated results from both local and external sites to establish baseline information on the absolute range of accuracies. The evaluation was carried out by classes based on scatter plots and the full set of validation metrics (Fig. 7).

In general, *Impervious surface* fractions yielded the most accurate results. MAE and RMSE varied between from 8.51 to 10.94 and from 13.99 to 17.23, respectively. Apart from *M-17's* prediction for 2017 with $R^2 = 0.89$, the rest exceeded 0.91. Slope saw consistent values, ranging from 1.11 to 1.16.

Woody fractions yielded slightly higher errors, with MAE and RMSE values ranging between from 11.03 to 12.11 and from 15.27 to 17.48, respectively. Compared to *Impervious surface*, the overall R^2 values decreased for *Woody*, ranging from 0.87 to 0.91. Highest slope value was witnessed in M-18's predictions with value 1.27, while others yielded around from 1.11 to 1.16.

Low vegetation showed the overall highest errors. MAE and RMSE values ranged from 12.12 to 13.66 and from 16.22 to 18.76, respectively. Particularly, R^2 values were higher than 0.89 in most years, with the lowest value of 0.87 witnessed in *M-17*.

3.2. Temporal and spatial transfer of standard models

We assessed the temporal transferability of each single-year model (standard models) to predict LC in years not represented in the training data (for information on annual data density of the training years and the predicted years see Fig. 2). Additionally, we separated the validation results from local and external areas to further assess the spatial transferability. Evaluations based on MAE are illustrated in Fig. 8. Additional information on RMSE, Slope, Intercept, R^2 and scatter plots are provided in Supplement 1.

From a model-wise perspective, *M-18* had the poorest temporal transferability, as mapping unseen years showed a large increase in MAE compared to its above-average performance in 2018. In particular, the increased errors were mainly related to the poor performance in estimating the *Woody* and *Low vegetation* fractions, MAE ranging from 15.13 to 18.31 and 15.07 to 18.48, respectively. In contrast, *Impervious surface* fractions showed lower and more consistent errors (MAE slightly below or around 10.0). Other models showed mixed performances when transferred to unseen years. For example, *M-17* gave better results when mapping other years compared to results achieved for the year it was trained in. Similarly, *M-21* had good transferability in most years (except 2017), with MAE often equal to or slightly lower than the baseline results. *M-19*, *20, 22* showed some increases in MAE when predicting other years, but not as high as *M-18*.

From a mapped-year perspective, all models performed best when applied to 2018, i.e., the year with the highest CSO (highest MAE for *Impervious surface*, *Woody* and *Low vegetation* were 11.52, 12.98 and 13.9, respectively). In contrast, all models produced the highest overall errors when applied to the year 2017, the year with the lowest CSO, (highest MAE for *Impervious surface*, *Woody* and *Low vegetation* were 11.89, 16.47 and 18.48 respectively).

When comparing the model performance on local and external sites, the differences were generally rather small with most models performing worse when evaluated on external sites. Only a few models showed a higher relative decrease, for example, M-20 predicting external areas of 2022, the MAE in *Low vegetation* increased by about 25% (from 11.86 to 14.88). In some cases, external sites showed higher accuracies. For example, M-22 predicting 2017, *Woody* yielded 10% less MAE when comparing results in local and external areas (13.87 and 12.46 respectively).

3.3. The impact of random observation selection on model performances

We evaluated the impact of ROS for all single-year models. Results were compared to the standard models, with the relative improvement (−%) or decrease (+%) of absolute MAE values (Fig. 9). Additional information on RMSE, Slope, Intercept, R^2 and scatter plots are provided in Supplement 2.

When assessing the performance of the same year mapping, ROS demonstrated slight improvements on local sites, ranging from −1% to −8%, for all three LC classes However, noticeable improvements were observed when maps were evaluated in external sites, with improvements in all LC types with up to −16% relative reduction of MAE for *Impervious surface* in case of M-18 predicting 2018 as well as M-20 predicting 2020 (9.36 reduced to 7.88 and 9.92 reduced to 8.36, respectively). For the rest, the MAE for the three LC types was constantly reduced by about −3% to −12% on external sites.

Mapping unseen years further demonstrates the improvements made by ROS. For instance, *M-18-R* improved substantially compared to the standard model, with relative reductions of MAE for *Impervious surface*, *Woody* and *Low vegetation* of up to −10%, −17%, and − 14% (local sites) and − 19%, −14%, and − 16% (external sites), respectively. Therefore,

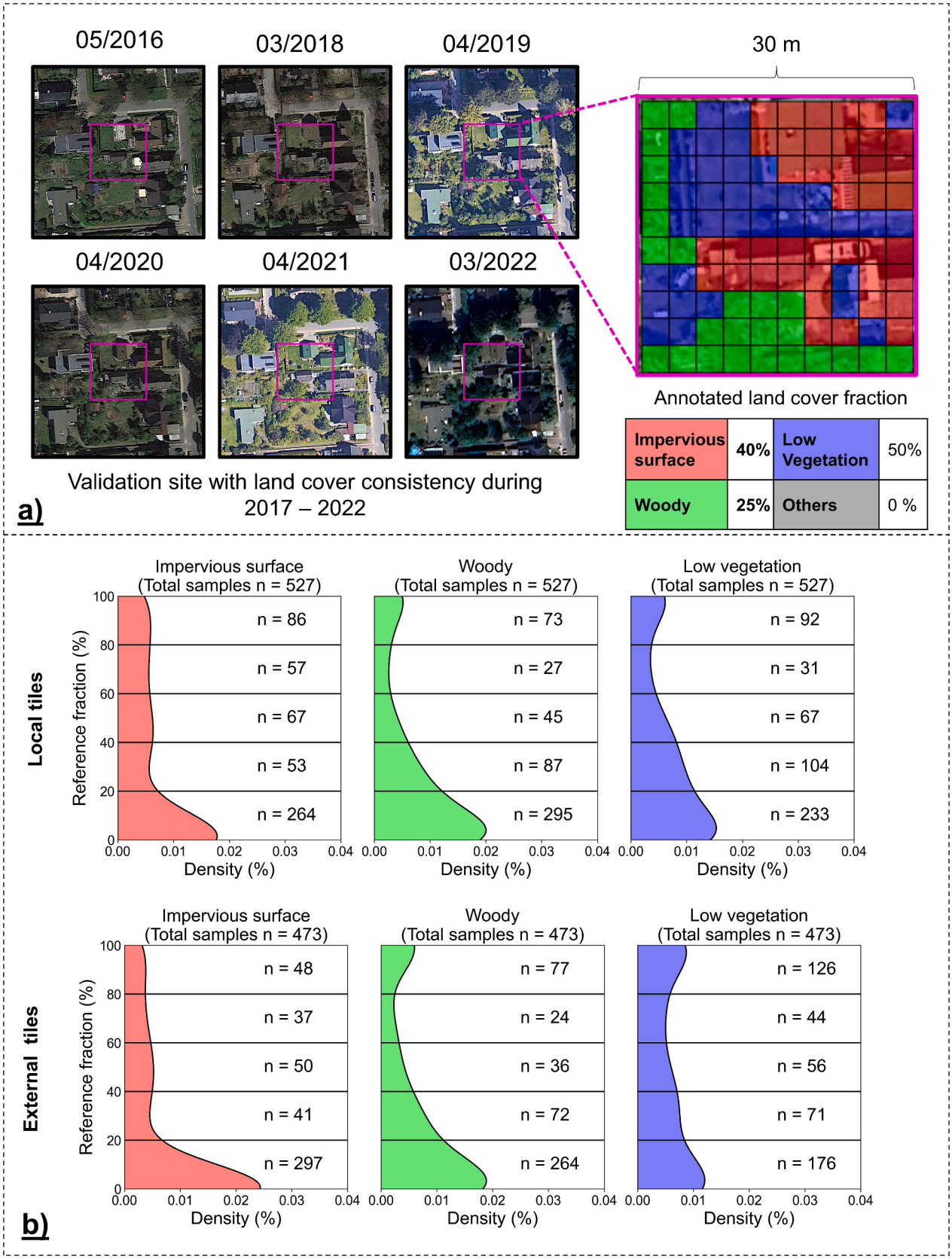

Fig. 6. a) Validation area example (30 × 30 m) VHR imagery proves temporal inconsistency during the period of 2017–2022 (left). A grid of 10 × 10 cells with 3 m resolution was overlaid on the area of 3 × 3 pixels and each cell was labelled with its dominant LC type. The fraction of each LC was calculated from the ratios of labels (right). b) Density distribution of validation data by land cover fraction values for local and external tiles and total count of validation points per 20% intervals.

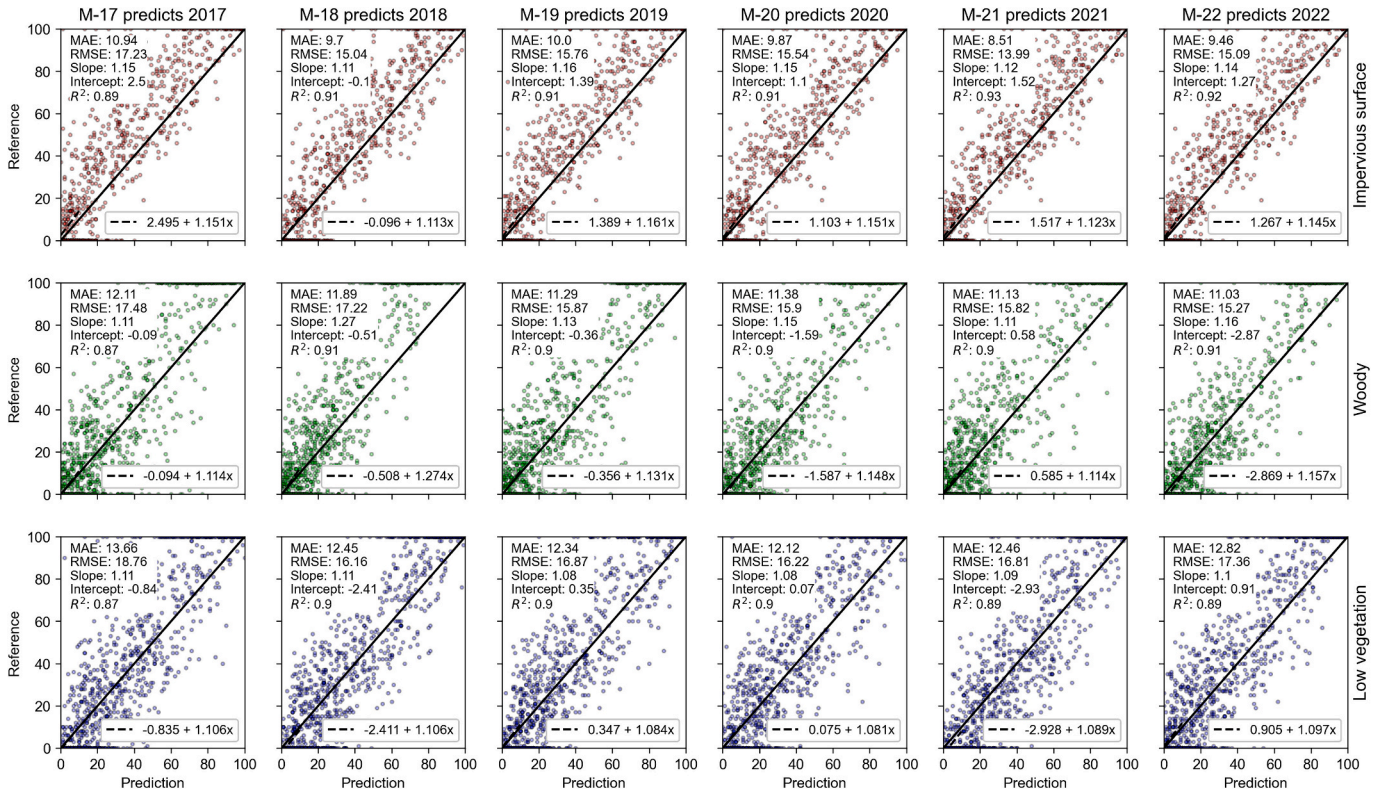

Fig. 7. Scatter plots and validation metrics for three LC (*Impervious surface*, *Woody* and *Low vegetation*) produced by the models trained on data from the same year.

the average relative improvement of model accuracies was −15% in terms of MAE. Improvements differed between models, e.g., *M-17-R* and *M-21-R* showed a reduced MAE for *Impervious surface* (about −12%) in one case (prediction 2020, external sites), while the errors for *Woody* and *Low vegetation* were relatively similar to the standard model in most cases. For other models, *M-(19, 20, 22)-R*, saw relative decreases of −3.2%, −7.5% and − 6.2% in average MAE when mapping in unseen years, respectively.

Looking at the improvements per mapped year, all models (except the same year model *M-17-R*) improved most after ROS when mapping 2017, the year with the lowest CSO. In contrast, ROS led to the smallest improvements when mapping 2018, the year with the highest CSO.

3.4. Performance of models trained on data from multiple years with and without random observation selection

We compared the results from the multi-year models (standard and ROS) with those from previously shown models, this time for class averages (MAE) as well as average accuracies integrated for the entire 6-year period (Fig. 10). Additional information on RMSE, Slope, Intercept, R^2 and scatter plots are provided in Supplement 3.

Overall, *M-ALL* outperformed all standard single-year models both for local and external sites. Thus, it also resulted in the lowest average MAE for the averaged 6-year predictions of 10.38 and 10.82, respectively. Similar to the single-year models, *M-ALL* produced the best results for 2018 and the worst for 2017. When applying ROS, *M-ALL-R* further improved for all years and both for local and external sites, resulting in the average MAE for the 6-year period decreasing from 10.38 to 10.17 in local sites and from 10.82 to 10.1 in external sites. Thus, accuracy for local and external sites were on average quite identical.

On a class-wise basis, the estimated *Impervious surface*, *Woody* and *Low vegetation* yielded MAEs of 9.56, 9.9, 11.05 in local sites and 7.91, 10.8, 11.61 in external sites respectively, with R^2 values >0.9 for all individual classes (Fig. 11).

3.5. Monitoring annual land cover fractions from 2017 to 2022

We mapped annual LC fractions (*Impervious surface, Woody, Low vegetation*) for northern Germany and Poland in the period 2017–2022 using the best generalized model (*M-ALL-R*) (Fig. 12a). The effectiveness of the model for monitoring LC fractions over time was investigated for areas with singular LC (pure pixels) and mixed LC (mixed pixels) both for changing as well as invariant surfaces.

An industrialization process example (Fig. 12a) demonstrates the model quality both regarding LC consistency and conversion. Between 2017 and 2019 no changes were observed. During the years 2020–2022, annual fraction maps capture the complete conversion of LC from forest to industrial areas. Specifically, the dense forest (shown as green in 2019) was cleared and left abandoned for an extended period, leading to the growth of low vegetation (blue color in 2020). Construction began in early 2021, resulting in a mixture of Impervious (red) and bare soil (black) backgrounds. By 2022 the soil area continued to expand and part of it converted into impervious surfaces. Similar processes can be seen in construction of a commercial building in Poland (Fig. 12b). A vast agricultural land, mapped as high fractions of *Low vegetation* in the first three years (2017–2019), displayed major change in LC fractions. Specifically, the beginning of construction work in 2020 with exposed bare soil revealed from the VHR image, was illustrated correctly from the fraction maps. In 2021 and 2022, the fraction maps showed the construction progress in each year, with building blocks (red) appearing on top of areas which were previously bare soil (black).

To further investigate areas with mixed pixels we analysed histograms of cover fractions in heterogeneous areas (Fig. 13). In a densely populated urban area in Berlin, Germany (Fig. 13a), where no LC changes occurred during this period, the model produced annual maps with consistent distributions of LC fractions. Here, the fraction maps displayed superiority in their capacity to capture both the predominant

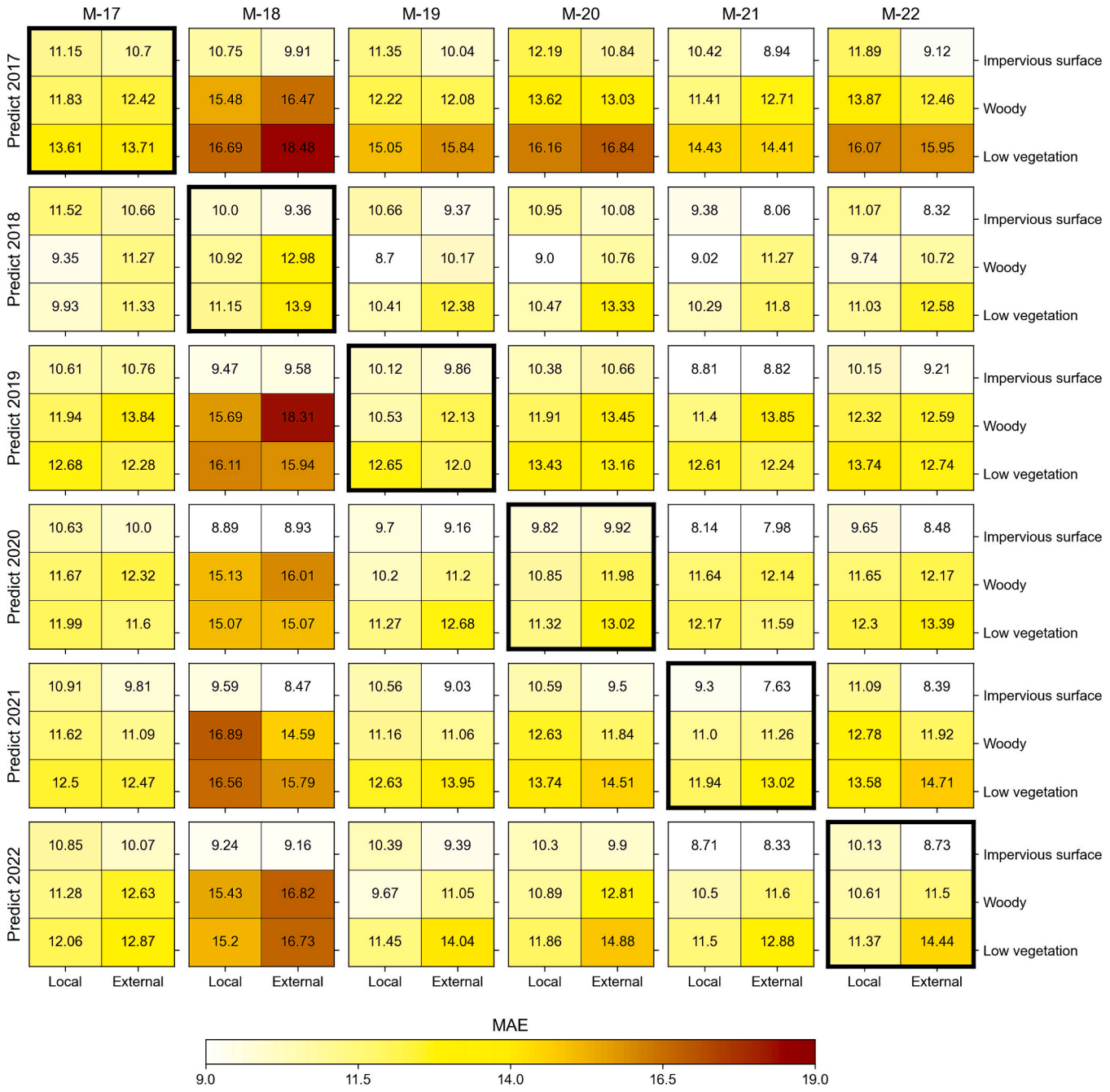

Fig. 8. Mean absolute errors (MAE) for standard models on local and external validation sites when applied to all years. The diagonal results (highlighted frames) showed the model's baseline performance in the same year.

class (*Impervious surface* fraction, which accounted for approximately 50% of the pixels) and sub-classes (*Woody* and *Low vegetation*, each accounting for major fractions ranging from 20 to 30%). In a more vegetation-dominated area (Fig. 13b), the heathland with scattered wooded trees displayed a high share of *Woody* fractions in most pixels, ranging from 20 to 60%. While the LC types remained relatively consistent over the given period, there were fluctuations in the LC fractions' distribution among three classes. Specifically, annual fractions of *Impervious surface* fluctuated greatly from 0 to 30%, whereas *Low vegetation* fractions were more consistent around 20 to 40%. In another example where LC change occurred, i.e., a field of solar panels was installed in an agricultural area in Germany (Fig. 13c). The panels were accurately mapped with the mixed pixels assigned to both *Impervious surface* and *Low vegetation*. For the first four years (2017–2020), the LC consisted of *Low vegetation* pixels (90–100% fraction). However, in 2021, a large area of solar panels was installed, resulting in an increase of mixtures between *Impervious surface* and *Low vegetation* in that year and the subsequent year. This correctly marks the new LC mix at sub-pixel level. During the first four years, a small area of solar panels to the north consistently appeared as mixed pixels in the histograms.

4. Discussion

4.1. The sensitivity of using annual STM and regression-based unmixing for quantifying land cover fractions for multiple years

We carried out a range of experiments using STM features derived from annual Sentinel-2 CSO (2017–2022) with regression-based

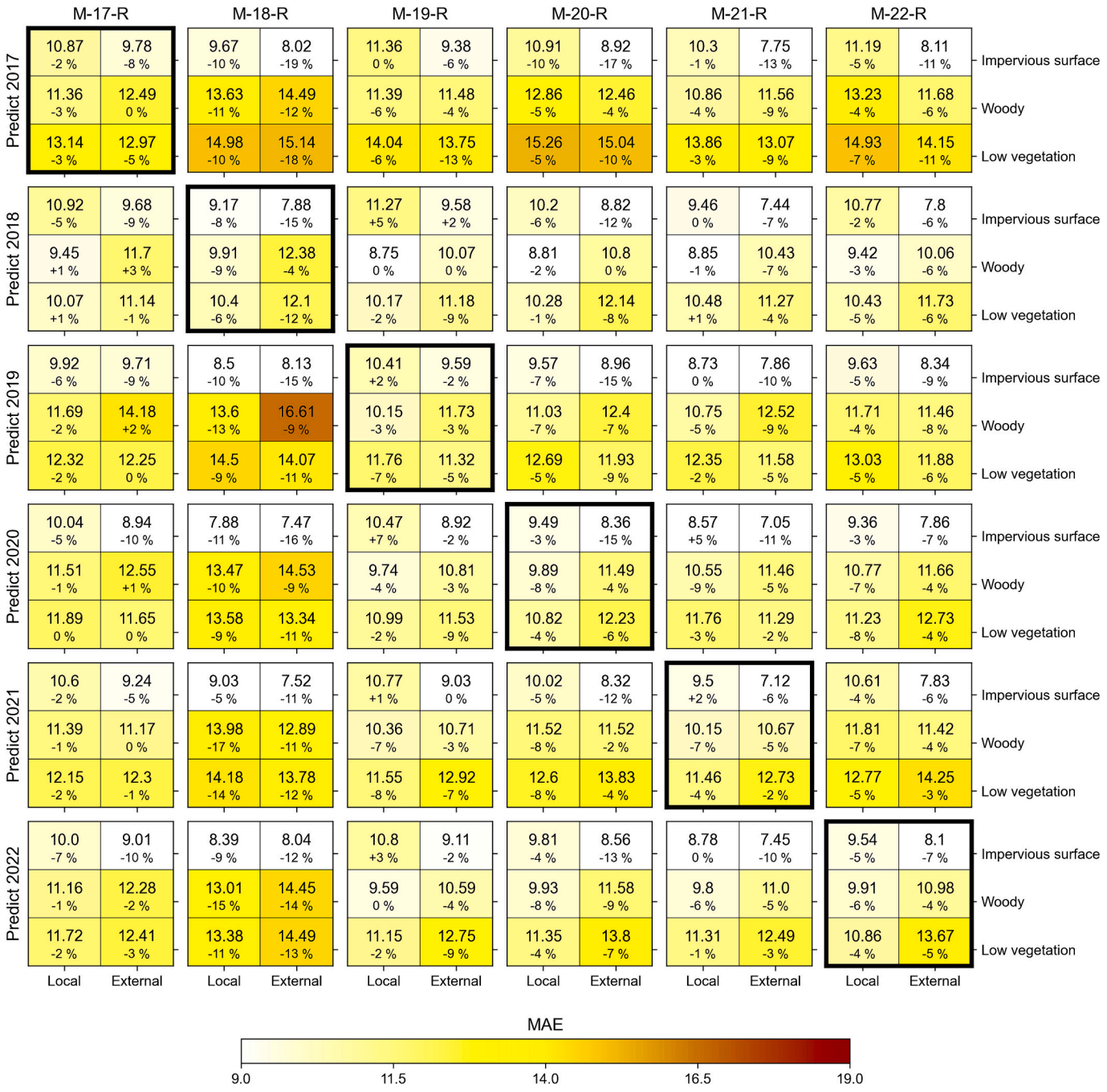

Fig. 9. Mean absolute errors (MAE) with relative increase (+%) and relative decrease (−%) when comparing ROS and standard models applied to all years for local and external validation sites. The diagonal results (highlighted frames) showed the model's baseline performance in the same year.

unmixing to evaluate the influence of annual data density on regression accuracy with a focus on model transfer to external validation sites and to unseen years.

We initially assessed the performance by training a model on STM profiles from a single year and applying it to the same year. This way, we first evaluated the quality of the workflow and of the training data and we got a baseline for all following comparisons. Mapping results were of high quality for all years and LC classes, where *Impervious surface* produced the lowest errors and most consistent results, followed by *Woody* and *Low vegetation* with slightly higher errors. The scatter plots and all metrics for the same year models on local validation areas show errors comparable to previous studies (Okujeni et al., 2018; Schug et al., 2020). The scatter plots show typical patterns of often underestimated higher fractions and overestimation of values near zero. Nevertheless, the MAE and RMSE with values below 12 and 18 demonstrate that the mapped information goes clearly beyond that of discrete classification, while keeping the effort for training data search similar to that of LC classification, i.e., only the labeling of pure pixels is required. Moreover, the STM proved principally useful for representing the annual variation between the different LC classes and the multi-class NNR proved a reliable approach.

Comparing the results between local and external areas, the performance in the external regions was only marginally worse. This spatially separate validation showed the ability of the annual models to accurately map larger areas within the same biome without a substantial decrease in quality. The observed decrease in accuracy for external areas

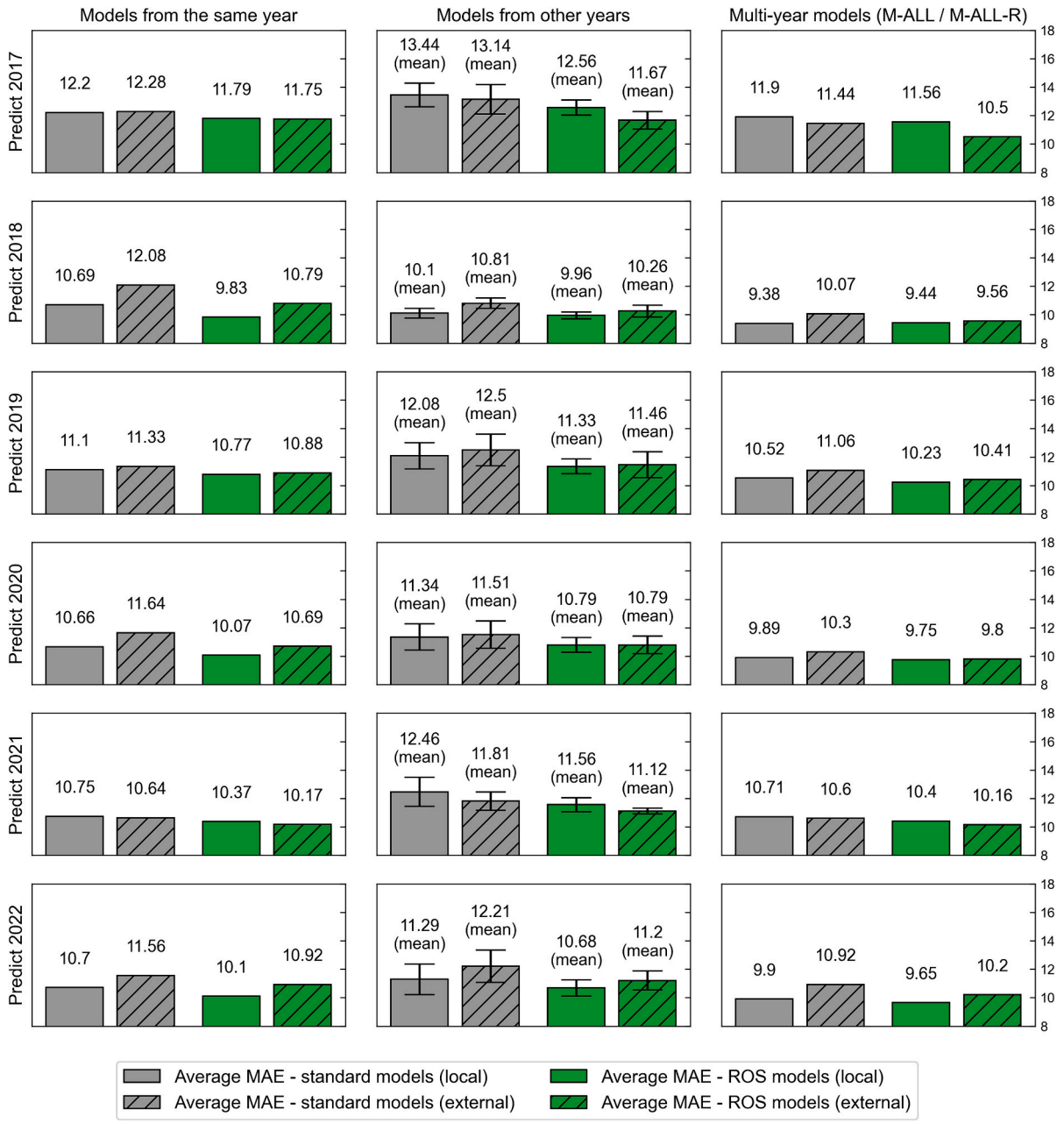

Fig. 10. Average MAE from two validation sites (local and external) for each year when mapped with single-year model from the same year (left columns) when mapped with single-year models from other years (centre columns) and when mapped with all-year model (right columns). Colors highlights the side-by-side comparison between standard (gray) and ROS (green) approaches. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

relates to the two vegetation classes while MAE for *Impervious surface* remains stable. We attribute this pattern to regional differences in phenology due to a more continental climate in Poland as well as to slight differences in temporal data density, which is smaller than inter-annual differences (Fig. 2). Nevertheless, even the highest error (*Low vegetation* MAE = 13.66 and RMSE = 18.76 in 2017) remained within an acceptable range and in line with earlier studies (Okujeni et al., 2018; Schug et al., 2020). This highlights the value of STM as continuous and relatively robust input features for regression-based unmixing of LC fractions over larger areas.

When mapping over multiple years, developing a model for each year individually can be deemed a reasonable strategy for steady LC monitoring. Nonetheless, this hinges on the availability of annual pure spectral data, or specifically, the presence of training data for each year. When transferring single-year models to map unseen years, we found that the annual CSO had a significant impact on both the performance of transferred models and the results achieved by all models on certain years. Our analysis revealed that a model trained in the year with highest CSO (*M-18*) had the lowest temporal transferability, resulting in poor performance when mapping years with sparser CSO. Conversely, results decreased less or even improved when a model was trained in a year with relatively low CSO data (e.g., *M-21*). This can be explained by the increasing stability of the STM with higher data density (Frantz et al., 2023), and hence, less variation learned during training. Models trained on STM from high data density years generalize worse. Models trained on low data density years, on the other hand, learn more variation and generalize better, when applied to other years. This influence of data density reverses when evaluating results by mapped years: maps

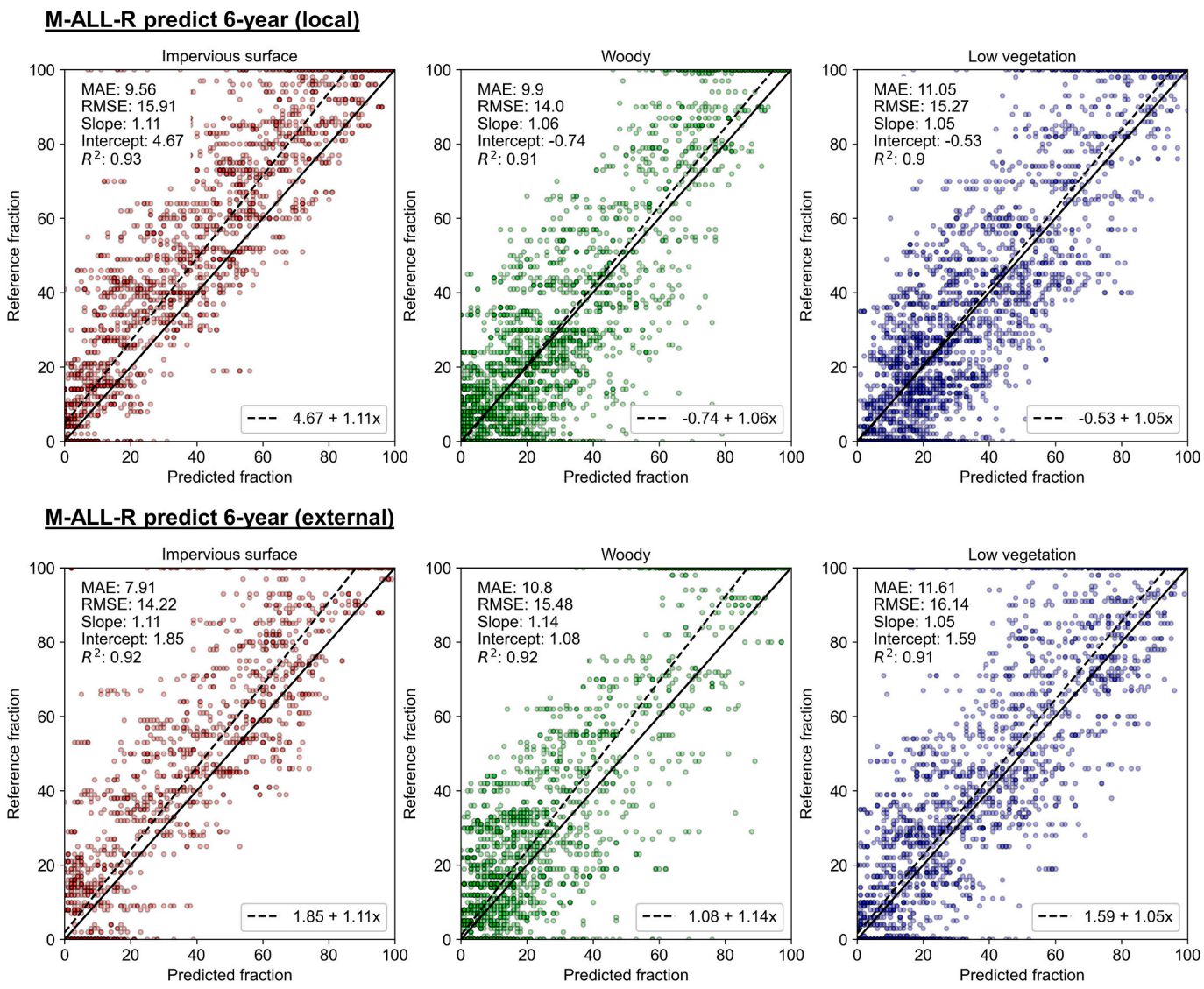

Fig. 11. Detailed mapping performances of the generalized model (M-ALL-R) for 6-year predictions. Scatter plots shown for individual LC fraction estimates (*Impervious surface, Woody* and *Low vegetation*) on local and external sites.

for data sparse years are worse because the high variation may cause some regions to be not well represented in different models. In high data density years, however, all regions can be expected to be represented by relatively typical STM and hence well known by all models. With regard to research question 1, we can state that not only does a high impact of differences in annual data density exist when mapping multiple years, the impact also has to be considered in a combination of model training and application. However, this is complicated, due to two aspects: all years are very different in weather conditions and the appearance of LC types. Hence, the model results and transferability are not only a function of data density but also of varying climate and hence land cover conditions. To have a more concrete answer for the impacts of data density on model results, we would need to have constant land cover conditions under varying data density. Such information would be part of another research avenue, which, e.g., could be answered with the approach proposed in Frantz et al. (2023).

The patterns of individual class accuracies support our explanation of the differences caused by data density. The temporally inconsistent, *Impervious surfaces* are characterized by little or no variation in the STM profile, and data density has little influence. Therefore, model transfer causes little variation in class-wise errors. The other two classes, *Woody* and *Low vegetation* are strongly influenced by phenology. In the later class, the mixing of vegetation and soil signals even changes the shape of the STM profiles, and missing periods of open soil or dense vegetation in sparse years thus have a very large impact on STM profile variation (Fig. 3). In this case, the impact of temporal distribution becomes relevant. For example, if annual CSO are mostly distributed in the middle of the growing season, STM will yield higher near-infrared values compared to other years. In contrast, when annual CSO are distributed within pre- and post-growing season (e.g., post harvesting crop area), the more CSOs with open soil result in high annual STM values in the shortwave-infrared. This further adds justification to our answer to research question 1 that there is a clear impact of temporal data density on regression-based unmixing with STM. This effect also hints to the drawback of using annual STM as input features; Annual STM aggregate information over time and high or low values cannot be related to a certain season or acquisition date. This way, temporal shifts in phenology can also not be reconstructed from STM. This will lead to underperformances when characterizing highly seasonal LC classes, e.g., during crop type classification (Rufin et al., 2019) or fraction mapping at species level (Arasumani et al., 2023). In consequence, annual STM can be considered strong for quantitative mapping of large areas. Yet this may be limited to LC types that differ in the magnitude of the used metrics, while LC types with temporally shifted phenologies or much

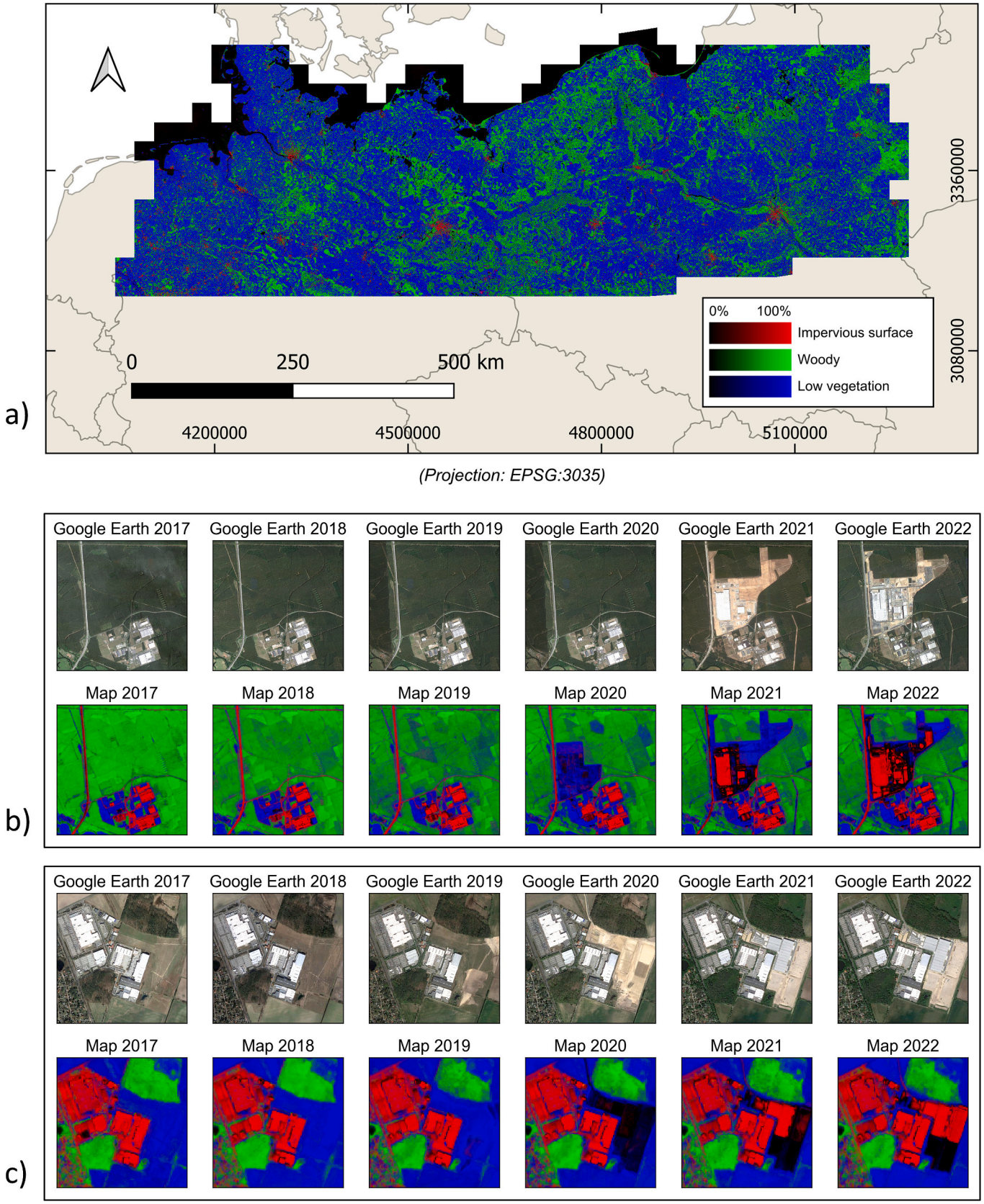

Fig. 12. a) Overview of fraction map for northern Germany - Poland (2022): *Impervious surface*, *Woody* and *Low vegetation* produced by the generalized model (M-ALL-R).). Maps for other years are provided in Supplement 4. b-c) Examples of gradual LC conversions in Germany (b) and Poland (c) during 2017–2022 captured by VHR images and the estimated fractions maps.

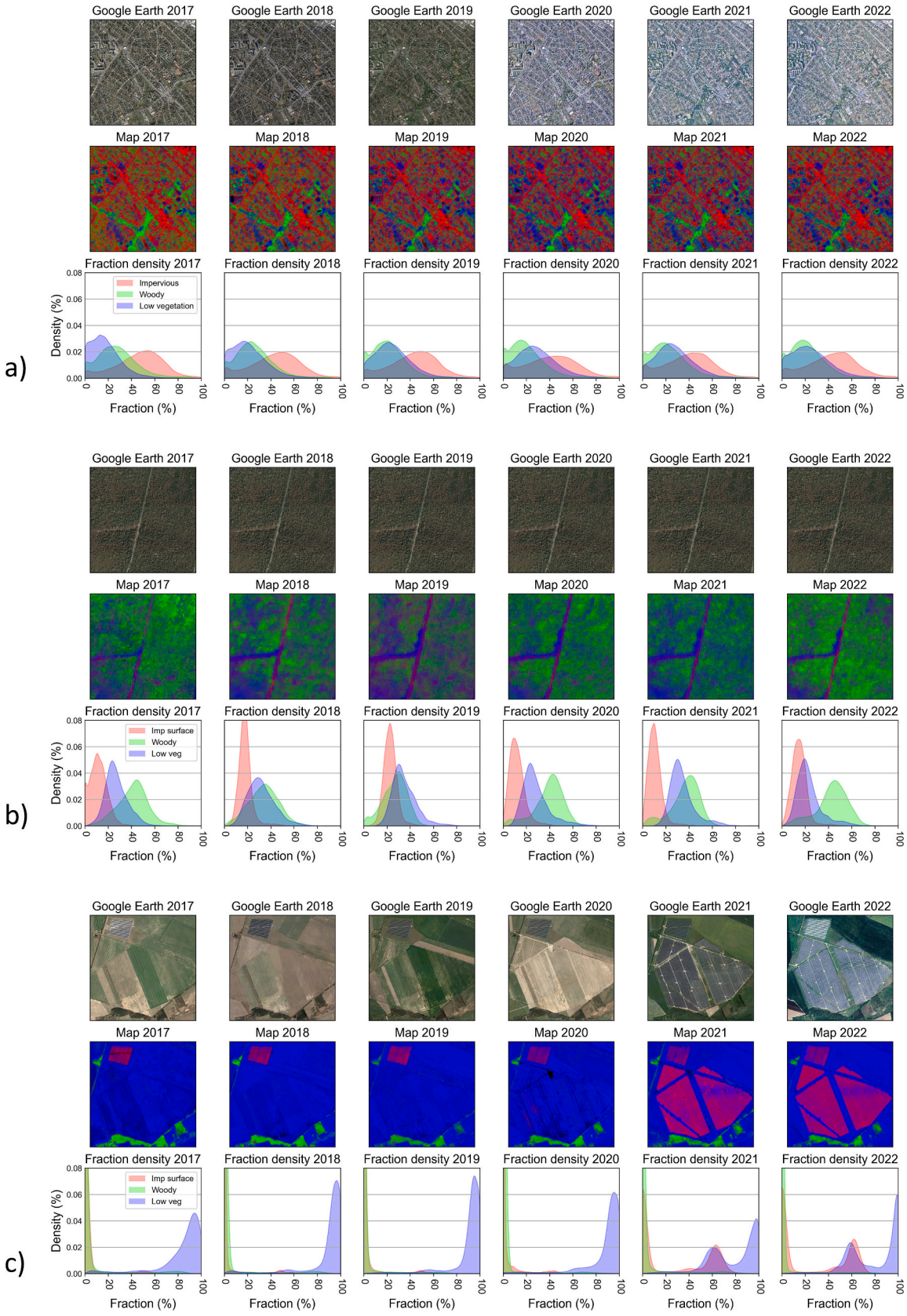

Fig. 13. Visual and quantitative LC fractions of heterogenous surfaces during 2017–2022. a) An unchanged urban area in Berlin – Germany; b) Heathland area in northern Poland; c) Solar panels installation in northern Germany.

higher numbers of different vegetation classes may not be mapped as accurately.

4.2. Random observation selection (ROS) and practical use

Results achieved with the standard models not only showed the impact of differences in data density during model training and applications, but also that more variation of STM improves model quality when transferred to other years. This can be achieved for single-year training with the proposed Random Observation Selection (ROS) to create STM to simulate such variations. Training with ROS led to improvements in mapping accuracies in almost all cases. When looking at results for spatial and temporal transfer in more detail, the findings achieved with the standard model are further supported. In the case of mapping the same year, ROS have marginal impact (slight negative impact in some cases) on local application, yet the previous deficiencies when mapping the external sites, i.e., spatial transfer within the same year, were further enhanced by ROS. We expect ROS to add the variation needed to account for the differences in data density between the western and eastern part of the study area. The fact that even local application is usually improved shows that already within the area of the six tiles used for training and validation, small differences in data density exist. However, our results showed that ROS had an even more positive impact on temporal transferability, especially for models that were trained in high data density years (i.e., *M-18*, *M-20*). Thus, the augmented STM are most valuable for models trained on data from years with high numbers of CSO and stable STM. Thus, the use of ROS was generally positive for single-year models, especially for models with weaknesses during temporal transfer. But also, for the best-performing model that was trained on data from all years (*M-ALL*) a positive effect of ROS can be observed. We explain this by the fact that the number of CSO and hence temporal density also vary throughout space (see Fig. 2), due to the orbit constellations; the smaller portion of the area lies in single orbits while the majority of the area is located in regions of orbit overlap. Hence, the higher temporal observation situation prevails in all models and a poorer representation of more sparse areas in standard models can be assumed. The data density of these areas is simulated and presumably improved by ROS. Hence, ROS not only improved the temporal transferability when dealing with different data density in the same areas (local validations), but also the spatial transferability (external validations) where spectral variations are also major challenging factors.

Although the implementation of ROS was shown not to harm the model performance, the computation cost is much higher in the training process compared to standard models. In our experiments, we augmented the original data four times with four selection rates of ROS (20%, 40%, 60%, 80%), thus resulting in 4-fold training time for each model. For a multi-year model with ROS (*M-ALL-R*), the training time can be 24 times longer (6 years, 4 augmented datasets) than a standard single-year model. Hence, although the answer to research question 2 is clearly positive and the advantage of ROS generally shown, the decision whether to include ROS in an analysis depends on three criteria that can be deducted from our experiments: (1) when reference information is available for all years that shall be mapped, a multi-year model can be expected to achieve very good results that already represent different data density and, hence, ROS may only slightly improve results; (2) when reference information is only available for a single year with data density lower than or similar to the years to predict, ROS will have a positive impact, yet training times should drive the decision for ROS; (3) when reference information is only available for a single year with above average data density, ROS will clearly improve results for lower temporal observation years and the extra processing time appears well justified.

4.3. Monitoring annual land cover fractions

In order to finally assess the value for a multi-year monitoring with a consistent framework, we mapped LC fractions for six consecutive years (2017–2022) based on annual STM of Sentinel-2 A/B data over northern Germany and Poland with the best generalizing model. The regression-based unmixing approach with synthetically mixed STM profiles yielded good accuracies for the three target classes of *Impervious surface*, *Woody* and *Low vegetation* and it also showed clearly, when the background class was mapped in areas with permanently open soil or water. The time-series of fraction maps with multiple classes were predicted using only one single NNR model, which marks a reduction in the complexity compared to the traditional class-specific regressors used in previous studies with similar approaches. Visual inspection of the resulting maps showed that neither artefacts could be observed, nor spatially related to CSO frequencies along orbit boundaries throughout the large area covered. Annual fraction maps have proven useful in capturing gradual LC conversions in areas where changes occurred. In our study area, we found that LC conversions, such as the construction of built-up areas, were often observed in the process of LC changing from *Low vegetation* to bare soil, and then to *Impervious surface*. More importantly, the major advantage of fraction maps is their ability to preserve sub-pixel information, especially when compared to categorical classification maps of highly heterogeneous surfaces. Quantitative information can reveal distinctive patterns of fraction distribution, which can serve as a more reliable indicator for identifying land use types or changes in land management. For instance, in our study areas, densely populated urban areas could be observed containing a mixture of all three LC types: *Impervious surface*, *Woody*, and *Low vegetation* (Fig. 13a). Conversely, other artificial infrastructure, such as solar panels, consisted of spatially homogeneous fraction mixture of *Impervious surface* and *Low vegetation* (Fig. 13c). However, there were inconsistencies when mapping invariant surfaces in more challenging heterogeneous areas, such as heathland areas with mixtures of grasslands and scattered trees (Fig. 13b). These areas yielded fluctuating annual estimates of LC fractions, likely due to the highly variable annual CSO and the unique LC characteristics that included a high proportion of tree shadows in the pixels. Therefore, there were clear limitations to the approach of mapping these areas, and the estimates of fractions may be more prone to errors than in other heterogeneous areas.

5. Conclusions

Our study delved into a critical aspect regarding the impact of variations in clear sky observations on annual land cover fraction monitoring with spectral-temporal metrics and regression-based unmixing. In the results, we identified clear patterns: (i) models trained on high data density years achieved better results when applied to the same years, but (ii) showed lower quality after temporal transfer. Models from lower data density years, on the other hand, were shown to (iii) be more robust when transferred to other years. Overall, accuracies achieved with (iv) models trained on reference information from multiple years were most robust and accurate. For the likely situation that reference information is only available for single (often recent) years, our suggested approach of (v) Random Observation Selection clearly improved the accuracy of temporally and spatially transferred models trained with data from a single year. The general improvement when including Random Observation Selection underlines the need for training data to incorporate spatial and temporal variation and thus for data augmentation approaches. For future work, we see a need to further explore the approach when used with more challenging land cover categories such as crop types, in regions with more challenging observation availability, or in constellations where data from different sensors are combined.

CRediT authorship contribution statement

Vu-Dong Pham: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Fabian Thiel:** Writing – review & editing. **David Frantz:** Writing – review & editing, Supervision. **Akpona Okujeni:** Writing – review & editing. **Franz Schug:** Writing – review & editing. **Sebastian van der Linden:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This research was conducted in the frame of the Interdisciplinary Research Center for the Baltic Sea Region Research (IFZO) of University of Greifswald, Germany, and the research project Fragmented Transformations, which is funded by the German Federal Ministry of Education and Research (FKZ 01UC2102). We thank the anonymous reviewers for their constructive comments and suggestions.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.rse.2024.114206.

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